Python: warm-up exercises on branching

This notebook was generated in pseudocode mode.

Exercise: Check if Number is Even

Exercise Description

Write a piece of code that checks whether a number `num` is even. If so, print a message.

Your solution

```
num = 42
# step 1: use the modulo operator % to check if num is divisible by 2
# step 2: if the remainder is 0, the number is even
# step 3: print a message informing that the number is even
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Even or Odd Message

Exercise Description

Write a piece of code that checks whether a number `num` is even. Print a message about the outcome (odd or even).

Your solution

```
num = 42
# step 1: use the modulo operator % to check if num is divisible by 2
# step 2: if the remainder is 0, the number is even, print even message
# step 3: otherwise, the number is odd, print odd message
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Conditional assignment

Exercise Description

Write a piece of code that checks whether a number `num` is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable `result`, depending on the outcome of the condition. Print the outcome at the end.

Your solution

```
num = 42
# step 1: create a variable to store the result (e.g., result = "")
# step 2: use the modulo operator % to check if num is divisible by 2
# step 3: if even, assign "even" to the result variable
# step 4: if odd, assign "odd" to the result variable
# step 5: print the result
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Multiple Divisibility Check

Exercise Description

Write a piece of code that checks whether a number `num` is divisible by 2, 3, 5 and 7. Print a message when `num` is divisible by one of this number.

Your solution

```
num = 42
# step 1: check if num is divisible by 2 using modulo operator, print message if true
# step 2: check if num is divisible by 3 using modulo operator, print message if true
# step 3: check if num is divisible by 5 using modulo operator, print message if true
# step 4: check if num is divisible by 7 using modulo operator, print message if true
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Nested Conditional Logic

Exercise Description

Write a piece of code that checks whether a number `num` is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

Your solution

```
num = 15
# step 1: check if num is even using modulo operator
# step 2: if even, print even message and check divisibility by 3
# step 3: if odd, print odd message and check divisibility by 5
# step 4: for each nested check, print appropriate message about divisibility
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Dog Age Calculator

Exercise Description

Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

Your solution

```
# step 1: ask the user for the dog's age using input() and convert to integer
# step 2: check if the age is 2 or less
# step 3: if age <= 2, multiply age by 10.5 to get dog years
# step 4: if age > 2, calculate as 2 * 10.5 + (age - 2) * 4
# step 5: print the result
```

Debug your solution